Lecture 8

Open-system dynamics: the GKSL equation

Objectives. *A channel describes one change of state. Modelling changes by a continuous, time-homogeneous semigroup imposes more structure. In finite dimension its generator has the GKSL form. We prove that this form generates channels, sketch why it is necessary, and derive the relation between population relaxation and coherence decay for a qubit.*

- Identify the semigroup and continuity assumptions, and prove that a given finite GKSL generator yields channels.
- Explain conditional Choi positivity in the converse sketch without claiming that the generator itself is CP.
- Solve the qubit Bloch equations and distinguish positivity from CP in the T_1, T_2 constraint.

8.1 A model with no remembered starting time

Throughout the main theorems $\mathcal{H} = \mathbb{C}^n$. A *quantum dynamical semigroup* is a family of channels Φ_t, $t \geq 0$, satisfying

$$\Phi_0 = \mathrm{id}, \qquad \Phi_{s+t} = \Phi_s \Phi_t, \qquad \|\Phi_t - \mathrm{id}\| \longrightarrow 0 \quad (t \downarrow 0).$$

The norm is an operator norm on the finite-dimensional matrix space; all such norms give the same convergence. Continuity at zero implies it at every time: small-time maps have a common finite norm bound, and the semigroup law bounds the maps on any finite time interval by splitting it into small steps. Then $\Phi_{t+h} - \Phi_t = \Phi_t(\Phi_h - \mathrm{id})$ and $\Phi_t - \Phi_{t-h} = \Phi_{t-h}(\Phi_h - \mathrm{id})$ give right and left norm continuity. Composition depends only on the elapsed times, rather than on when the experiment starts. The same rule applies to every intermediate state without retaining system–environment correlations or environmental memory as extra input. There is no requirement that Φ_t have a positive inverse, and in general negative times are not channels.

A continuous semigroup on a finite-dimensional vector space has a bounded linear generator $\mathcal{L}$ with $\Phi_t = e^{t\mathcal{L}}$ and $\dot{\rho} = \mathcal{L}(\rho)$ (cited: Rivas–Huelga, Theorem 1.3.1 pp. 5–6 and Corollary 1.3.1 p. 7). This exponential uses a matrix power series on the vector space of matrices. The Hamiltonian exponential acts on vectors instead; the two spaces and their generators differ. Our forward proof starts with a specified $\mathcal{L}$ and proves its channel property directly, without using this cited converse-to-continuity step as an input.

L7's decay channels illustrate the semigroup condition. Their excited population multiplier is $a(t) = 1 - \gamma(t)$, and composition multiplies $a(s)a(t)$. Taking $a(t) = e^{-t/T_1}$ supplies the semigroup law; an arbitrary $\gamma(t)$ need not. For a fixed initially separate environment, each

map $\Phi_{t,0}(\rho) = \mathrm{Tr}_E[U_t(\rho \otimes \sigma_E)U_t^\dagger]$ is a channel. At an intermediate time s, the joint state may be correlated, and L7's correlation example shows why the marginal need not determine its future. A semigroup additionally posits a channel $\Phi_{t,s} = \Phi_{t-s}$ satisfying $\rho(t) = \Phi_{t,s}(\rho(s))$. Factorisation is not automatically restored.

Self-check 8.1

1. Is a family of fixed-time channels automatically a semigroup?
2. On which vector space does $e^{t\mathcal{L}}$ act?

8.2 Why GKSL form produces channels

For matrices A, B, write $\{A, B\} = AB + BA$.

Theorem 8.1 (Finite GKSL, forward direction). *Let $H = H^\dagger$ be an $n \times n$ matrix and let $L_1, \ldots, L_d$ be a finite list of matrices. Then*

$$\mathcal{L}(X) = -\frac{i}{\hbar}[H, X] + \sum_{k=1}^{d}\left(L_k X L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, X\}\right) \tag{8.1}$$

generates a semigroup of channels $\Phi_t = e^{t\mathcal{L}}$ for $t \geq 0$.

Proof. Set $R = \sum_k L_k^\dagger L_k$ and $A_0 = -iH/\hbar - R/2$. For a time step $h \geq 0$ the map

$$\Psi_h(X) = (\mathbb{1} + hA_0)X(\mathbb{1} + hA_0)^\dagger + h\sum_k L_k X L_k^\dagger$$

is CP by its Kraus form. Expanding gives exactly $\Psi_h = \mathrm{id} + h\mathcal{L} + h^2\mathcal{R}_2$, where $\mathcal{R}_2(X) = A_0 X A_0^\dagger$. This step need not be TP: its Kraus completeness sum is $\mathbb{1} + h^2 A_0^\dagger A_0$. We use it only to approximate a CP limit.

Choose a submultiplicative operator norm for maps on the matrix space. The exponential power series and the exact expansion imply, for small h,

$$\left\|\Psi_h - e^{h\mathcal{L}}\right\| \leq c_1 h^2, \qquad \|\Psi_h\|, \ \left\|e^{h\mathcal{L}}\right\| \leq e^{Mh}$$

with fixed finite constants c_1, M. For example, $\|\Psi_h\| \leq 1 + h\|\mathcal{L}\| + h^2\|\mathcal{R}_2\|$; the exponential remainder is bounded by $h^2\|\mathcal{L}\|^2 e^{h\|\mathcal{L}\|}/2$. For any two maps A, B, the telescoping identity $A^N - B^N = \sum_{j=0}^{N-1} A^{N-1-j}(A - B)B^j$ holds without commutation. Taking $h = t/N$ therefore gives

$$\left\|\Psi_{t/N}^N - e^{t\mathcal{L}}\right\| \leq N c_1 (t/N)^2 e^{Mt} = \frac{c_1 t^2 e^{Mt}}{N} \longrightarrow 0.$$

Every finite composition is CP: products of its Kraus operators form a Kraus list. Its Choi matrix is positive by L7; entrywise convergence passes every nonnegative quadratic form to the limit, so the limiting Choi matrix is positive. L7 implies $e^{t\mathcal{L}}$ is CP.

Independently, finite trace cyclicity in (8.1) gives $\mathrm{Tr}\,\mathcal{L}(X) = 0$ for every X. Differentiating the convergent power series yields $\frac{d}{dt}\mathrm{Tr}(e^{t\mathcal{L}}X) = \mathrm{Tr}\,\mathcal{L}(e^{t\mathcal{L}}X) = 0$. The trace is therefore its initial value, so the limit is TP. The exponential has the semigroup law and norm continuity by its power series, completing the proof. ∎

The approximation does not interpret every short step as a normalised physical experiment. Trace preservation is proved for the exact limit, not borrowed from the approximants. Nor

does the proof assume that a matrix generator is positive or CP. It is the exponential that must map states to states, just as a Hamiltonian itself is not an evolution operator.

The L_k are called *Lindblad operators*; each has units time$^{-1/2}$. Their list and the split into Hamiltonian and noise terms are not unique. A different representation can describe the same generator, so identifying an individual term with a measured event requires an additional experimental model. The finite theorem proves no statement about arbitrary unbounded L_k on L^2.

Self-check 8.2

1. Why are the maps Ψ_h CP but generally not channels?
2. Which estimate ensures that their N-fold product approaches the exponential?

Example 8.2 (One generator, an explicit channel)**.** For $H = 0$ and $L_1 = \sqrt{\kappa/2}\sigma_z$, $\kappa \geq 0$, the generator is $\mathcal{L}(X) = \kappa(\sigma_z X \sigma_z - X)/2$. Diagonal entries are constant and off-diagonal entries acquire the factor $e^{-\kappa t}$. Thus its exact solution is

$$\Phi_t(X) = \frac{1 + e^{-\kappa t}}{2} X + \frac{1 - e^{-\kappa t}}{2} \sigma_z X \sigma_z.$$

Both coefficients are nonnegative and sum to one, so L7 independently recognises the solution as a random-unitary channel. It leaves the two σ_z rays fixed, while suppressing their relative coherence. Multiplying the off-diagonal factors verifies the semigroup law directly. The same formula can describe classical phase noise or an ignored quantum record. The reduced state alone does not determine which environment produced it, a distinction needed for the measurement problem in L12. If κ were negative, the off-diagonal factor would exceed one and $|+x\rangle\langle+x|$ would leave the Bloch ball. The nonnegative-rate restriction is visible here without solving a general matrix exponential.

8.3 Why every finite channel semigroup has this form

The converse is *sketched*; see Rivas–Huelga, Theorem 4.2.1 pp. 35–39, with constant coefficients for a semigroup. The source treats more general time-dependent divisible dynamics as well. Here the generator is time independent and the space finite dimensional.

Use L7's unnormalised $\Omega = \sum_i |ii\rangle$ and its Choi convention. For a differentiable channel family,

$$C_{\Phi_t} = |\Omega\rangle\langle\Omega| + tC_{\mathcal{L}} + o(t).$$

Let $E_\perp = \mathbb{1} - |\Omega\rangle\langle\Omega|/n$. For $v \perp \Omega$, positivity for $t > 0$ implies $0 \leq \langle v, C_{\Phi_t} v\rangle = t\,\langle v, C_{\mathcal{L}} v\rangle + o(t)$. Dividing by t and taking the limit yields

$$E_\perp C_{\mathcal{L}} E_\perp \geq 0. \tag{8.2}$$

It does not give $C_{\mathcal{L}} \geq 0$. The nonnegative part lives on the complement of the maximally entangled vector.

The Choi derivative $C_{\mathcal{L}}$ is Hermitian because each C_{Φ_t} is. Factor that compressed matrix by finite spectral theory. Reshape its vectors as in L7 to get a CP map $\mathcal{N}(X) = \sum_k L_k X L_k^\dagger$ whose Choi matrix is exactly the compression. The remaining Hermitian Choi matrix has zero compression onto $\Omega^\perp$. In a basis beginning with $\Omega/\sqrt{n}$, it has only a first row and column. It can therefore be written $|g\rangle\langle\Omega| + |\Omega\rangle\langle g|$ for a vector g. The first-row/column

assertion can be checked directly. Write the residual matrix as D in the decomposition $\mathbb{C}^n \otimes \mathbb{C}^n = \mathrm{span}\{\Omega/\sqrt{n}\} \oplus \Omega^\perp$. For a real scalar d and a vector $v \in \Omega^\perp$, it has block form

$$D = \begin{pmatrix} d & v^\dagger \\ v & 0 \end{pmatrix}, \qquad g = \frac{d}{2n}\Omega + \frac{1}{\sqrt{n}}v.$$

The coordinates of Ω are $(\sqrt{n}, 0)$ in this decomposition. Multiplying the two rank-one matrices gives

$$|g\rangle\langle\Omega| + |\Omega\rangle\langle g| = \begin{pmatrix} d/2 & 0 \\ v & 0 \end{pmatrix} + \begin{pmatrix} d/2 & v^\dagger \\ 0 & 0 \end{pmatrix} = D.$$

Reshape by $g = (\mathbb{1} \otimes A_0)\Omega$, so $g_{(i,a)} = (A_0)_{ai}$. Indeed $\sum_{ij} e_{ij} \otimes A_0 e_{ij} = |g\rangle\langle\Omega|$, and the adjoint expression gives $|\Omega\rangle\langle g|$. Thus the residual map is $A_0 X + X A_0^\dagger$. This is the algebraic step underlying the converse; the source gives the full coefficient argument.

Differentiating $\mathrm{Tr}\,\Phi_t(X) = \mathrm{Tr}\,X$ at zero gives $\mathrm{Tr}\,\mathcal{L}(X) = 0$ for every X, forcing the resulting decomposition to satisfy $A_0 + A_0^\dagger = -\sum_k L_k^\dagger L_k = -R$. Its anti-self-adjoint part has the form $-iH/\hbar$ with $H = H^\dagger$, so $A_0 = -R/2 - iH/\hbar$. Substitution gives (8.1). Together with the forward theorem, this describes all continuous finite channel semigroups. The sketch is a finite-matrix statement; it does not establish a general unbounded infinite-dimensional GKSL theorem.

Self-check 8.3

1. Why does positivity of the small-time Choi matrix give only the compressed condition?
2. Which part of A_0 supplies the Hamiltonian?

8.4 The damped qubit and its two times

Use $|0\rangle$ excited, $|1\rangle$ ground, and

$$H = \frac{\hbar\nu_0}{2}\sigma_z, \qquad L_1 = \frac{|1\rangle\langle 0|}{\sqrt{T_1}}, \qquad L_2 = \sqrt{\frac{\gamma_\phi}{2}}\sigma_z, \quad T_1 > 0, \quad \gamma_\phi \geq 0.$$

The first noise term loses excited population. The second leaves populations fixed but changes the sign of an off-diagonal entry inside $\sigma_z \rho \sigma_z$, so its net contribution there is $-\gamma_\phi \rho_{01}$. For $\rho = \begin{pmatrix} a & c \\ \bar{c} & 1-a \end{pmatrix}$ the two dissipators are

$$\mathcal{N}_1(\rho) = \frac{1}{T_1}\begin{pmatrix} -a & -c/2 \\ -\bar{c}/2 & a \end{pmatrix}, \qquad \mathcal{N}_2(\rho) = \begin{pmatrix} 0 & -\gamma_\phi c \\ -\gamma_\phi \bar{c} & 0 \end{pmatrix}.$$

The commutator contributes $-i\nu_0 c$ in the upper off-diagonal entry and its conjugate below. Adding these matrices gives

$$\dot{\rho}_{00} = -\rho_{00}/T_1, \tag{8.3}$$

$$\dot{\rho}_{01} = -\left(i\nu_0 + \frac{1}{2T_1} + \gamma_\phi\right)\rho_{01}. \tag{8.4}$$

Trace and Hermiticity give the other two entries. Define T_2 by the coherence decay rate:

$$\frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\phi, \qquad T_2 \leq 2T_1. \tag{8.5}$$

T_1 and T_2 are *times*; γ_ϕ is a rate. The factor $1/2$ in the pure-dephasing Lindblad operator is needed to produce precisely this rate, rather than twice it.

For initial $|+x\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, integration gives

$$\rho_{00}(t) = \tfrac{1}{2}e^{-t/T_1}, \qquad \rho_{01}(t) = \tfrac{1}{2}e^{-t/T_2}e^{-i\nu_0 t}.$$

With QM-I's $\rho = (\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})/2$, this is

$$r_x = e^{-t/T_2}\cos(\nu_0 t), \quad r_y = e^{-t/T_2}\sin(\nu_0 t), \quad r_z = e^{-t/T_1} - 1. \tag{8.6}$$

The Bloch vector spirals toward the ground pole $r_z = -1$. The probability of finding $+x$ is $[1 + e^{-t/T_2}\cos(\nu_0 t)]/2$, whereas the excited population is $e^{-t/T_1}/2$. These different observables separate the two times. In a frame rotating at ν_0 the oscillation is removed and the transverse radius simply decays. If $\gamma_\phi = 0$, L7's emission model is recovered and $T_2 = 2T_1$; extra dephasing shortens T_2 without altering the population decay.

The decay times can be compared without the free rotation. Divide the excited population by its initial value and the coherence magnitude by its initial magnitude. For this input the resulting curves are $a = e^{-t/T_1}$ and $b = e^{-t/T_2}$. Figure 8.1 compares pure emission with an additional dephasing rate. The population curve stays fixed while the coherence curve changes.

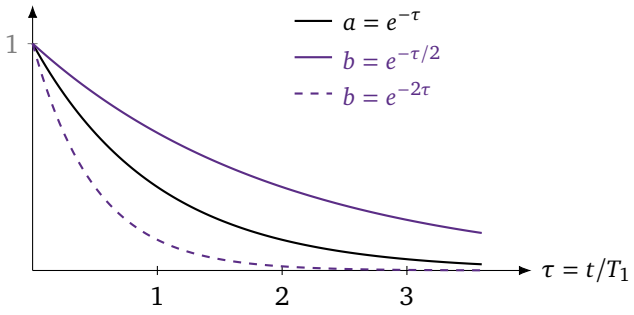


Figure 8.1: Normalised population a and coherence magnitude b for the initial $+x$ state. Solid violet gives $T_2 = 2T_1$; dashed violet gives $T_2 = T_1/2$, while the black population curve is the same in both models.

The physics

Population readings after inversion determine T_1 in this model; equatorial preparation and Ramsey or free-induction readings determine T_2 from the decaying transverse signal. These are distinct experimental procedures even when fitted by one Markov model. A cavity provides another open-system setting: Brune et al., *PRL* **77**, 4887 (1996), studied progressive decoherence of a mesoscopic microwave-field superposition. Their reservoir had a small nonzero thermal photon population, about 0.05; a vacuum-reservoir model is an approximation. That experiment is not a direct measurement of the qubit relation (8.5). The oscillator attenuation computation, with unbounded annihilation operator, is a separate Sheet 8 example outside this finite theorem's scope.

Self-check 8.4

1. Solve the population and coherence equations from $|+y\rangle = (|0\rangle + i\,|1\rangle)/\sqrt{2}$ and give the Bloch vector.
2. Which sign of the equilibrium r_z follows from the excited-state convention?

8.5 Where complete positivity adds a constraint

Consider independently specified population and coherence multipliers for relaxation to the pure ground state: the population is multiplied by a and the coherence by b. Positivity alone then imposes $T_2 \le 2T_1$ when they have the exponential parametrisation below. To see this, remove the irrelevant phase rotation, write $a = e^{-t/T_1}$, $b = e^{-t/T_2}$, and start with a pure state having excited population ϵ and coherence $\sqrt{\epsilon(1-\epsilon)}$. The output determinant is

$$\epsilon(a - b^2) + \epsilon^2(b^2 - a^2).$$

Nonnegativity for arbitrarily small $\epsilon > 0$ requires $b^2 \le a$. Thus this model alone cannot demonstrate the difference between positive and completely positive dynamics.

Use instead the *unital* Bloch map $(r_x, r_y, r_z) \mapsto (br_x, br_y, ar_z)$, with $0 \le a, b \le 1$. It describes the symmetry of equal upward and downward relaxation rates, with a maximally mixed equilibrium (an infinite-temperature idealisation) when the semigroup is CP. Independently of CP, the displayed map contracts the Bloch ball and is therefore positive. Its Choi matrix is

$$C = \frac{1}{2}\begin{pmatrix} 1+a & 0 & 0 & 2b \\ 0 & 1-a & 0 & 0 \\ 0 & 0 & 1-a & 0 \\ 2b & 0 & 0 & 1+a \end{pmatrix}.$$

Its eigenvalues are $(1-a)/2$ twice and $(1+a)/2 \pm b$. Complete positivity requires $2b \le 1+a$. With the exponential a, b above, a small-time expansion of the lowest eigenvalue gives

$$\frac{1+a}{2} - b = t\left(\frac{1}{T_2} - \frac{1}{2T_1}\right) + O(t^2).$$

If $T_2 > 2T_1$, these maps form a positive continuous semigroup but fail CP at small positive times. Conversely $T_2 \le 2T_1$ gives $b \le \sqrt{a} \le (1+a)/2$, hence CP at every time. The extra constraint has now been demonstrated in a model where ordinary positivity does not already impose it.

For a numerical spectator test, take $a = 0.8$, $b = 0.95$. Every isolated Bloch vector contracts, yet the lowest Choi eigenvalue is $(1+0.8)/2 - 0.95 = -0.05$. Acting on half of the normalised maximally entangled input produces $C/2$, with negative eigenvalue -0.025 along $(|00\rangle - |11\rangle)/\sqrt{2}$. A putative reading of that projection would have a negative probability. Testing only population and coherence bounds on single-system states would have missed it.

Finally, a semigroup is an assumption about the physical regime. Born, Markov and often secular or coarse-graining approximations enter microscopic reservoir derivations; none is proved by writing a GKSL expression. In a weak-coupling Born approximation, the reservoir state is kept close to a specified reference state when computing the reduced evolution. The Markov approximation neglects the dependence of the reduced rate on earlier system states over the reservoir's memory time. A secular approximation averages terms oscillating at sufficiently different transition frequencies. Coarse-graining averages over a time window short compared with relaxation but long enough to suppress the reservoir correlations being neglected. These are separate assumptions about coupling strengths and timescales. QM-II's short-time law (Proposition 12.8) requires a unit vector in $D(H)$; its cited Khalfin theorem (Theorem 12.9) also requires a lower energy bound. In QM-II's Friedrichs decay model the excited population is exactly that closed-system survival probability: the initial excited vector is the only excited-system sector. It starts quadratically, rather than with a nonzero

exponential slope, and cannot remain exponentially bounded at arbitrarily late times. Thus e^{-t/T_1} is an effective intermediate-time model there, with $1/T_1 \approx \Gamma$ from the cited golden rule (QM-II, Theorem 12.13). A generic reduced population is not automatically a survival probability; its relation to the closed dynamics must be checked, not used to evade the decay model's short- and long-time restrictions. For infinite systems later lectures build their channels explicitly; any differential generator statements have their own domain qualifications.

Self-check 8.5

1. Can a map preserve the Bloch ball yet make a joint system–spectator state nonpositive?
2. Why is the exponential population law only effective in the Friedrichs decay model?

How does an open system evolve in time?

A finite-dimensional open system modelled by a continuous time-homogeneous semigroup of channels obeys $\dot\rho = \mathcal{L}(\rho)$ with a GKSL generator. The forward implication is proved here and the converse sketched. The semigroup is a physical modelling assumption. Closed evolution is the case with no Lindblad terms. Infinite-dimensional generators require additional hypotheses beyond this finite theorem.

Summary. CP short steps converge in norm to the finite GKSL exponential; trace preservation is checked separately. Conditional Choi positivity explains the converse structure. Population and coherence decays give $1/T_2 = 1/(2T_1) + \gamma_\phi$. The unital qubit shows why CP is stronger than positivity. Exact subsystem semigroup behaviour and exact closed-system survival are different claims.

Exercises. Try Sheet 8 in the separate *Exercise sheets* booklet (English).

Self-check answers.

8.1 1. No; their composition law need not hold. 2. The vector space of matrices, not the system vector space.

8.2 1. Their Kraus sum is $\mathbb{1} + h^2 A_0^\dagger A_0$ rather than $\mathbb{1}$. 2. The telescoping bound $c_1 t^2 e^{Mt}/N$.

8.3 1. Only vectors perpendicular to Ω have zero constant term. 2. The anti-self-adjoint part, $-iH/\hbar$.

8.4 1. $\mathbf{r}(t) = (-e^{-t/T_2}\sin\nu_0 t, e^{-t/T_2}\cos\nu_0 t, e^{-t/T_1} - 1)$. 2. Negative: ground is $|1\rangle$ and has $r_z = -1$.

8.5 1. Yes; the positive unital semigroup with $T_2 > 2T_1$ supplies it. 2. That population equals survival there: the short-time derivative is zero, and the lower-bound Khalfin restriction excludes an exact exponential late-time law.

Sources. Rivas–Huelga, *Open Quantum Systems*, pp. 5–7, Theorem 4.2.1 pp. 35–39; Weinberg, *Lectures on Quantum Mechanics*, §6.9 pp. 237–243; Nielsen–Chuang, p. 386 (their Lindblad normalisation differs by $\sqrt{2}$); Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, pp. 155–167. The forward proof includes its own finite product estimate.